

DATA SCIENCE CAPSTONE PROJECT

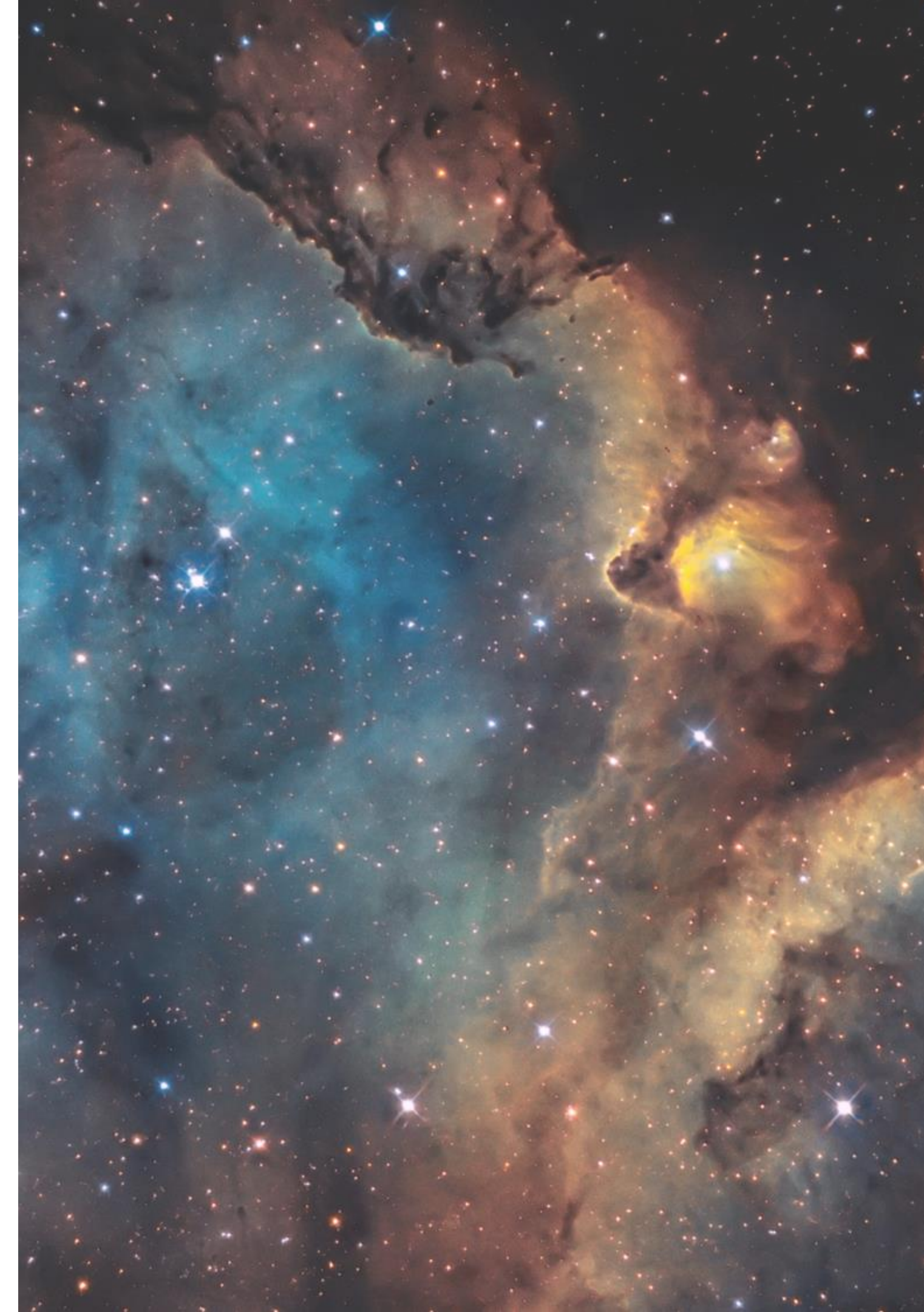
ANKE DE FREY

09/11/2024

[HTTPS://GITHUB.COM/ANKEDEFREY/DATA-SCIENCE-CAPSTONE-PROJECT](https://github.com/AnkeDeFrey/data-science-capstone-project)

OUTLINE

- P3** Executive Summary
- P4** Introduction
- P5** Methodology
- P17** Insights drawn from EDA
- P34** Launch sites proximities analysis
- P39** Dashboard with plotly dash
- P43** Predictive analysis (classification)
- P46** Conclusion
- P47** Appendix



EXECUTIVE SUMMARY

- Data was gathered from SpaceX API and the SpaceX Wikipedia page
- A new column “class” was created to label landings as either successful or not.
- The dataset was explored through SQL queries, visualization, folium maps and dashboards, with key columns chosen as features.
- Categorical variables were converted to binary format with one-hot encoding and the data was standardized.
- GridSearchCV was applied to fine tune the parameters of the machine learning models, and accuracy scored for each model were displayed.
- the analysis included four machine learning models: Logistic regression, Support Vector machine, Decision tree Classifier and K-Nearest Neighbors.
- All models achieved a similar accuracy of 83.33% but tended to overpredict successful landings
- Increasing the dataset would help enhance model performance and accuracy



INTRODUCTION

Project background and context

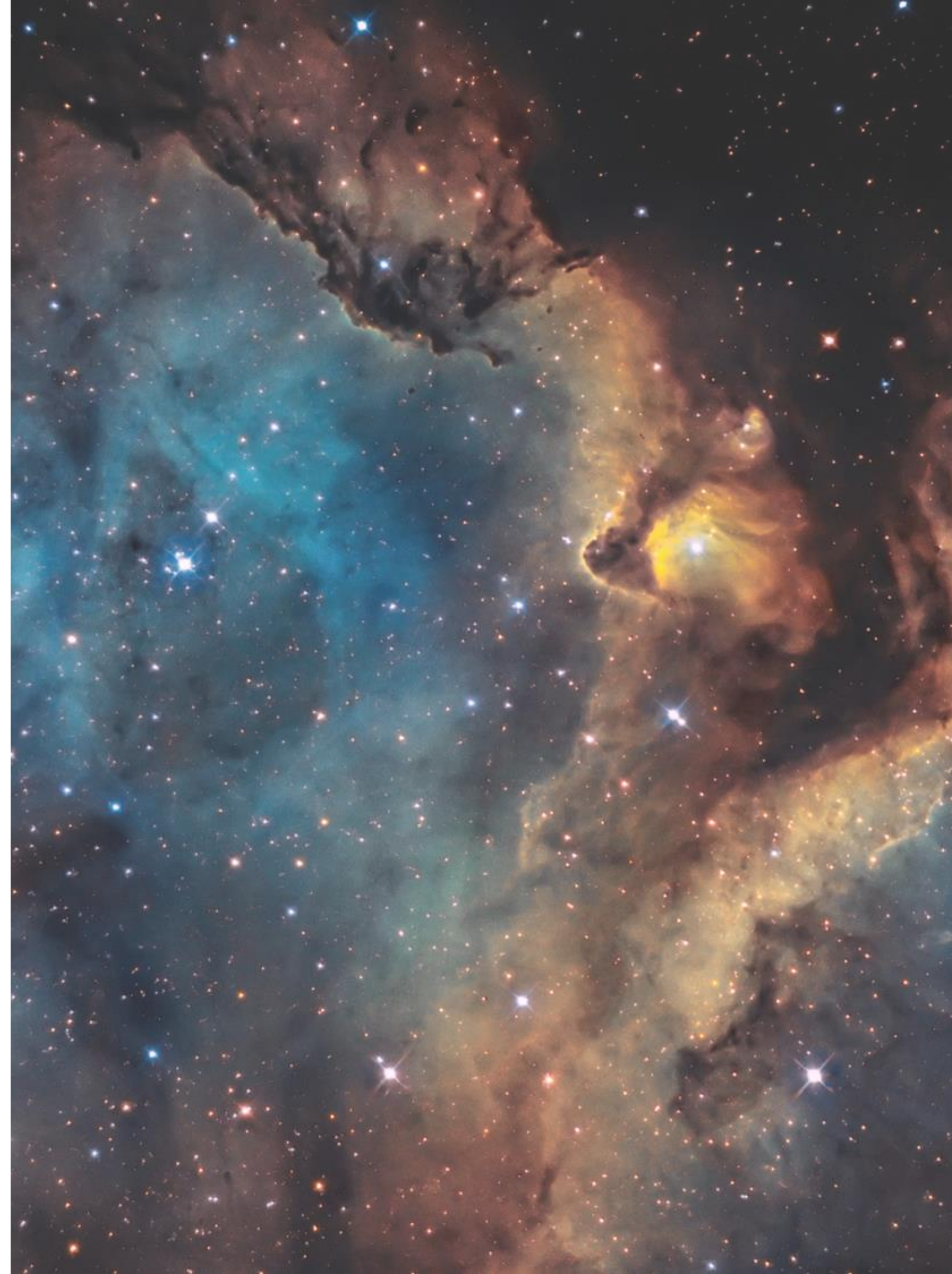
- Main objective: To Predict if the Falcon 9 first stage will land successfully
- Context: SpaceX offers relatively inexpensive rocket launches (\$62 million) compared to other providers (\$165 million+), largely due to reusability of the first stage.
- Data Collected on Space X's Falcon 9 launches via an API.
- The prediction model will help SpaceY, a new rocket company, determine competitive launch pricing.

Problems you want to find answers

- Predicting first stage landing success
 - Develop a machine learning model to predict whether the Falcon 9 first stage will land successfully based on historical data and launch parameters (payload mass, launch site, number of flights and orbits)
- Rate of successful landing over time
- Best predictive model for successful landing



METHODOLOGY



METHODOLOGY

Summary

- **Data collection methodology:** Via SpaceX REST API and web scraping techniques.
- **Perform data wrangling:** Filtering data, handling missing values and applying one hot encoding
- Perform **exploratory data analysis** (EDA) using visualization and SQL
- Perform interactive **visual analytics** using Folium and Plotly Dash
- Perform **predictive analysis** using classification models: How to build, tune, evaluate classification models



METHODOLOGY

DATA COLLECTION

- **SpaceX REST API:**

- api.spacexdata.com/v4/launches/past
- Provides detailed information about past SpaceX launches, such as the **rocket used**, **payload delivered**, **launch specifications**, **landing specifications** and the **outcome of the landing**.

- **Web Scraping (BeautifulSoup):**

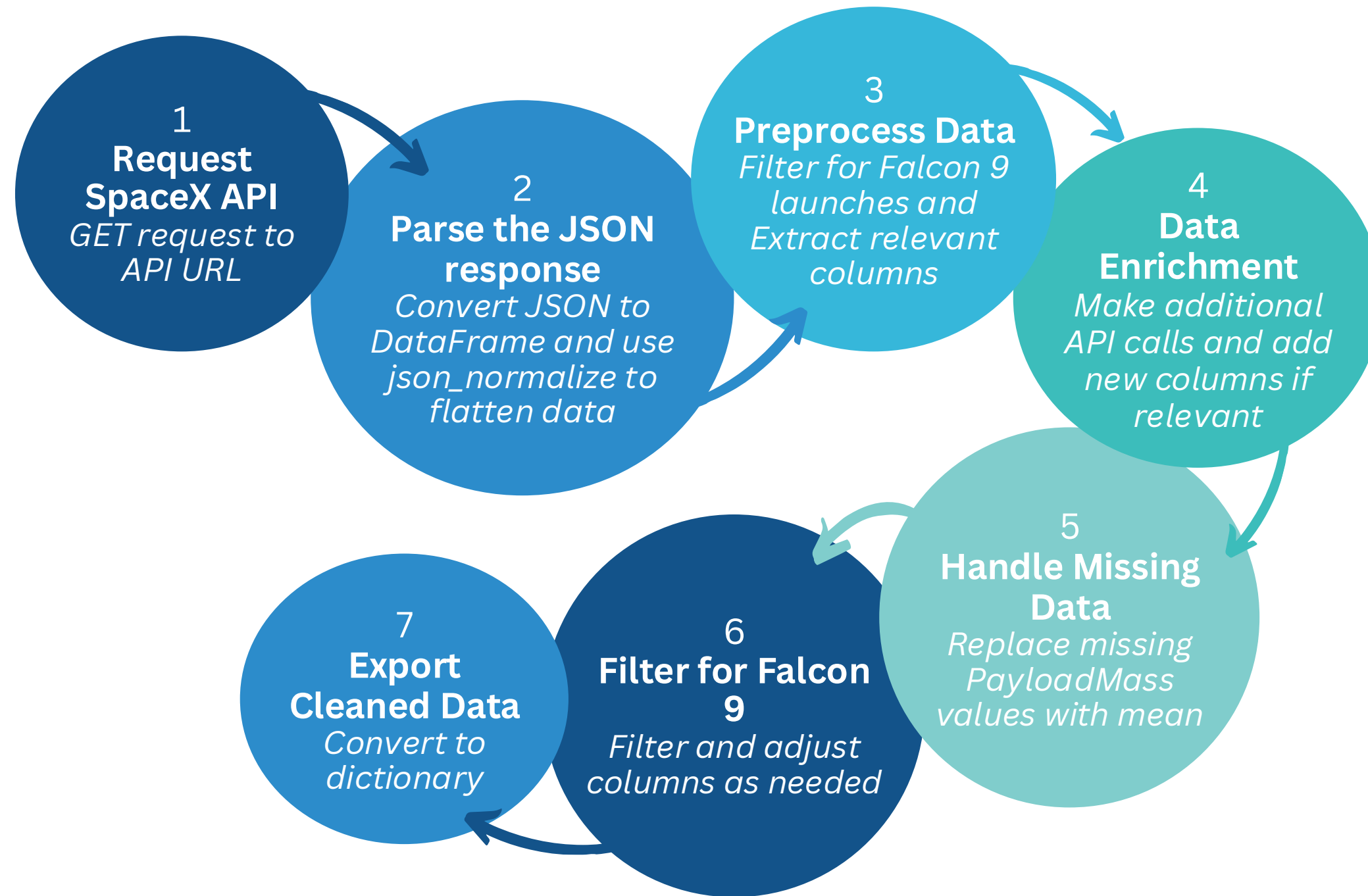
- Alternative data source obtained through **web scraping of HTML tables** from related **Wikipedia pages**.
- **Beautiful soup** used to **parse HTML tables** to **extract relevant launch data**.
- Once data is cleaned it is transformed into a **Pandas DataFrame** for further **analysis and visualisation**.

(see flowcharts on next page)



METHODOLOGY

DATA COLLECTION - SPACEX API

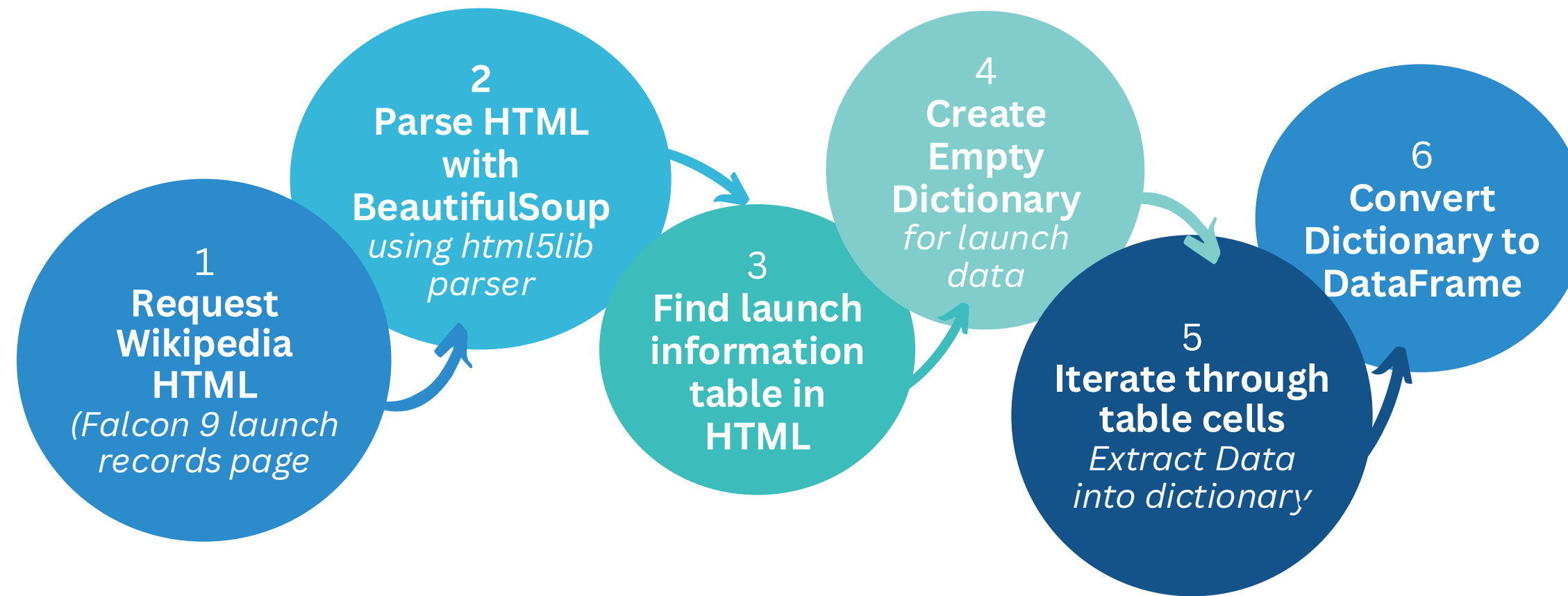


[https://github.com/Ankedefrey/
Data-Science-Capstone-
Project/blob/main/jupyter-labs-
spacex-data-collection-
api.ipynb](https://github.com/Ankedefrey/Data-Science-Capstone-Project/blob/main/jupyter-labs-spacex-data-collection-api.ipynb)



METHODOLOGY

DATA COLLECTION - WEB SCRAPING



[https://github.com/Ankedefrey/
Data-Science-Capstone-
Project/blob/main/jupyter-labs-
webscraping%20\(1\).ipynb](https://github.com/Ankedefrey/Data-Science-Capstone-Project/blob/main/jupyter-labs-webscraping%20(1).ipynb)



METHODOLOGY

DATA WRANGLING

- **Perform Exploratory Data Analysis (EDA)** and identify key labels for training data.
- **Calculate:**
 - **Number of Launches per Site:** Analyze launch frequencies by location.
 - **Number and Occurrence of Each Orbit Type:** Determine common orbit types and their frequency.
 - **Number and Outcome of Each Mission by Orbit Type:** Summarize mission outcomes per orbit.
- **Create Binary Landing Outcome Column:** Generate a dependent variable labeled Class to indicate landing success.
 - True Ocean: Successful landing in a designated ocean region.
 - False Ocean: Unsuccessful landing in an ocean region.
 - True RTLS: Successful landing on a ground pad.
 - False RTLS: Unsuccessful landing on a ground pad.
 - True ASDS: Successful landing on a drone ship.
 - False ASDS: Unsuccessful landing on a drone ship.
- **Convert Outcomes to Binary:** Assign 1 for successful outcomes and 0 for unsuccessful ones in the Class column.
- **Export Data to CSV:** Save processed data for modeling.

[https://github.com/Ankedefrey/
Data-Science-Capstone-
Project/blob/main/labs-jupyter-
spacex-Data%20wrangling.ipynb](https://github.com/Ankedefrey/Data-Science-Capstone-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb)



METHODOLOGY

EDA WITH DATA VISUALIZATION

[https://github.com/Ankedefrey/
Data-Science-Capstone-
Project/blob/main/edadataviz.ip
ynb](https://github.com/Ankedefrey/Data-Science-Capstone-Project/blob/main/edadataviz.ipynb)

Scatter plot

Shows relationships and trends between two variables (eg. flight number vs payload mass, launch site, orbit type or payload mass vs orbit type.



Scatter plot

- Flight Number vs Payload Mass
- Flight Number and Launch Site
- Payload Mass and Orbit type
- Flight Number and Orbit type

Bar chart

Compares categorical data, such as success rates for each orbit type.



Bar chart

- Success rate of each Orbit type

Line chart

Tracks changes over time, like the yearly trend of launch success.



Line chart

- Launch success yearly trend



METHODOLOGY

EDA WITH SQL

Queries

- Display unique launch sites in the space mission
- Display 5 records where launch sites begin with 'CCA'
- Calculate the total payload mass carries by boosters launched by NASA (CRS)
- Calculate the average payload mass carried by booster version F9 v1.1
- List the date of the first successful landing outcome in ground pad
- List boosters with successful drone ship landing and payload mass between 4000 and 6000
- List the total number of successful and failed mission outcomes
- List boosters carrying the maximum payload mass using a subquery
- Display failure landing outcomes in drone ship for the ear 2015, with month names
- Ranked landing outcomes by count between June4, 2010, and March 20, 2017, in descending order

[https://github.com/Ankedefrey/
Data-Science-Capstone-
Project/blob/main/jupyter-labs-
eda-sql-
coursera_sqlite%20\(1\).ipynb](https://github.com/Ankedefrey/Data-Science-Capstone-Project/blob/main/jupyter-labs-eda-sql-coursera_sqlite%20(1).ipynb)



METHODOLOGY

MAP WITH FOLIUM

https://github.com/Ankedefrey/Data-Science-Capstone-Project/blob/main/lab_jupyter_launch_site_location.ipynb

Markers for launch sites

- Added **folium.Circle** and **folium.Marker**.
- These objects provide a visual representation of where each SpaceX launch site is located, making it easier for users to identify launch site locations on the map. A popup label on each circle displays the site's name for context.

Markers for Launch Success/Failure

- Added **folium.Marker** with custom color icons, grouped using **MarkerCluster**.
- These markers show the success (green) or failure (red) of each launch. Cluster grouping is used to manage overlapping markers when there are many launches at the same location, enhancing the map's readability.

Proximity analysis object (Lines and Distance calculations)

- Added **folium.PolyLine** and **folium.Marker** for displaying the distance from launch sites to nearby features like coastlines, cities, railways and highways.
- The line connections between the launch site and these features are drawn to illustrate potential impact on the site on the selection and operations.

Distance Calculation Function

- Used **calculate_distance** to compute the distances between two geographical points using the Haversine formula.
- This calculation informs the proximity analysis and helps quantify the relationship between launch sites and nearby features

Mouse Position

- Added **MousePosition** widget.
- Provides real-time coordinates as you hover the mouse over the map, aiding in precise measurement and location identification during exploratory analysis



METHODOLOGY

DASHBOARD WITH PLOTLY DASH

Dropdown for Launch Site Selection

- Allows user to select a specific launch site or view data for all sites
- Provides a flexible way to filter data based on site, helping users focus on particular locations

Pie chart for launch success

- Displays the total number of successful launches, either across all sites or for a specific site.
- It visualizes launch success vs failure, helping users assess performance by site.

Slider for Payload Range

- Allows users to select a range of payloads.
- Provides interactivity to filter data by payload mass, enabling users to explore how payload affects launch success.

Scatter plot for Payload vs Success

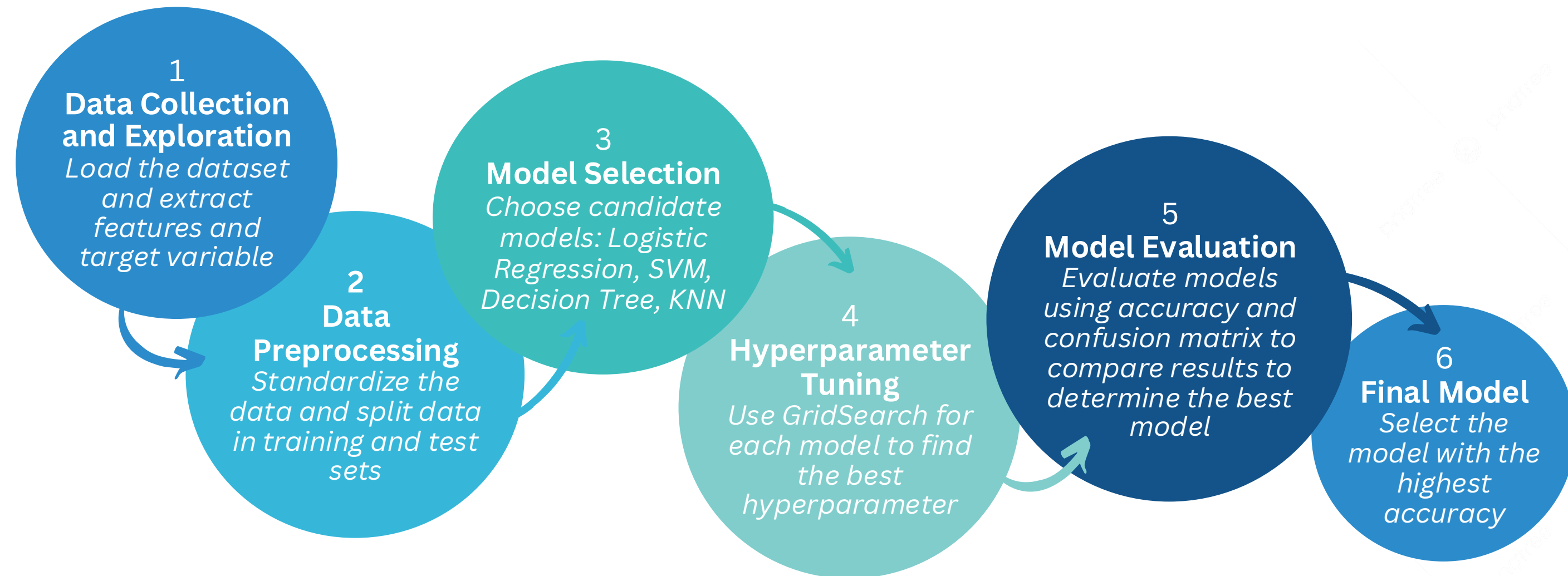
- Shows the correlation between payload mass and launch success, with different booster versions.
- Helps users analyze how payload size influences the likelihood of a successful launch, providing insights into the impact of payloads on success rates.

[https://github.com/Ankedefrey/
Data-Science-Capstone-
Project/blob/main/spacex_dash_
app%20\(1\).py](https://github.com/Ankedefrey/Data-Science-Capstone-Project/blob/main/spacex_dash_app%20(1).py)



METHODOLOGY

PREDICTIVE ANALYTICS



[https://github.com/Ankedefrey/
Data-Science-Capstone-
Project/blob/main/SpaceX_Mac
hine%20Learning%20Prediction
_Part_5.ipynb](https://github.com/Ankedefrey/Data-Science-Capstone-Project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb)



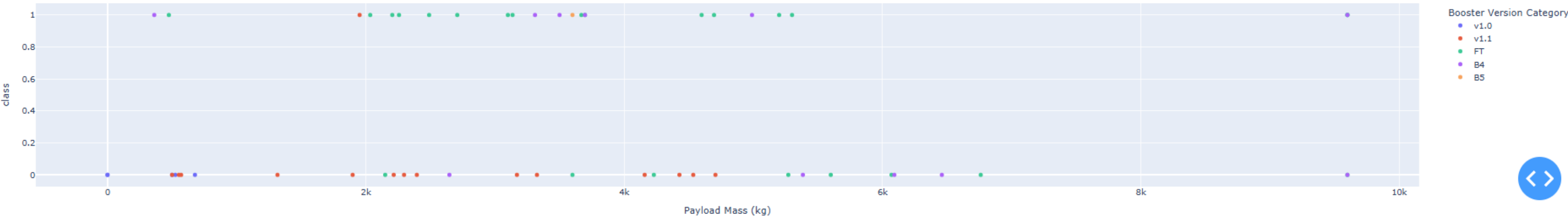
SpaceX Launch Records Dashboard

All Sites ✕ ▼

Total Success Launches by Site

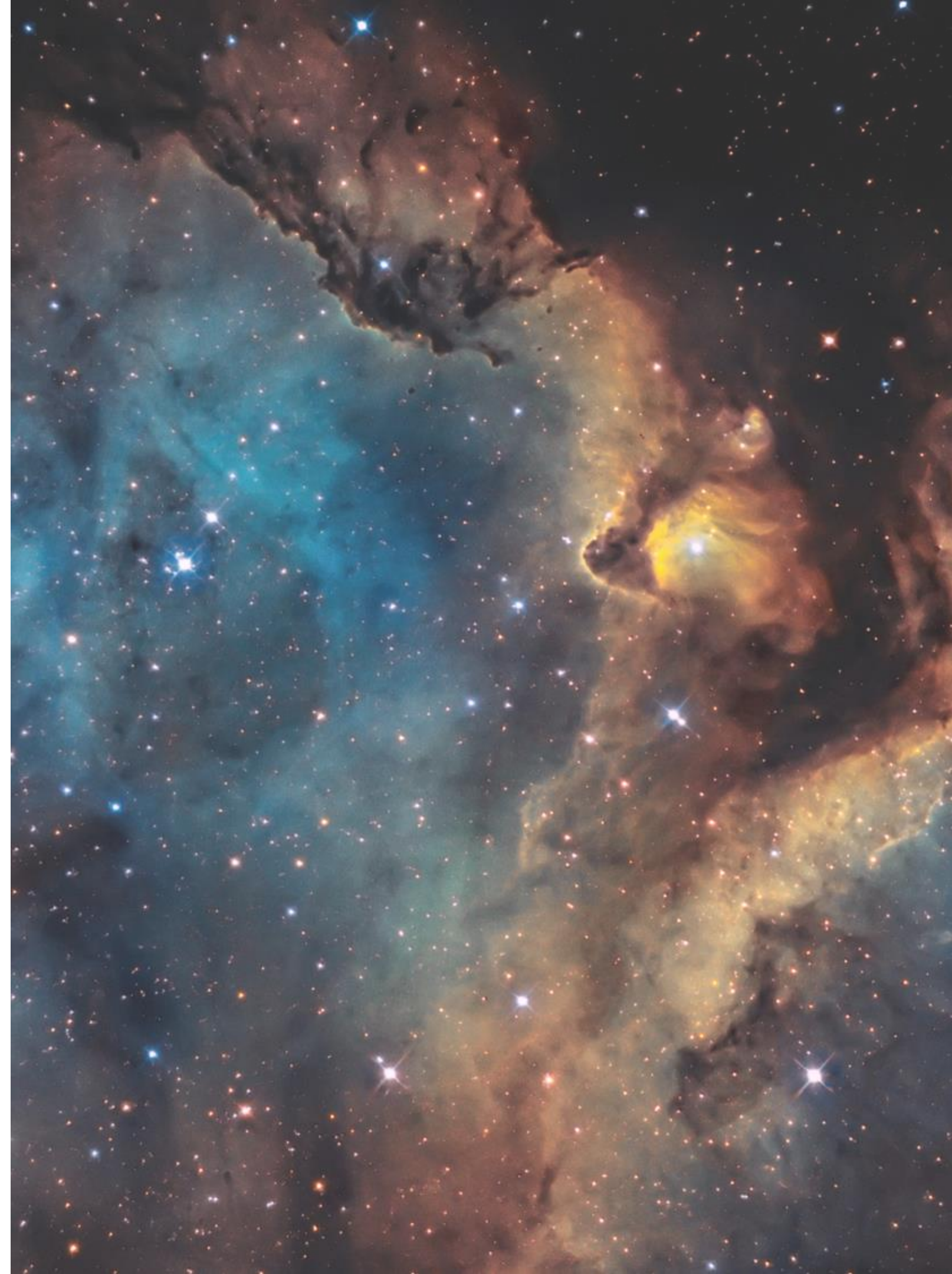


Payload vs. Outcome for All Sites



INSIGHTS DRAWN FROM EDA

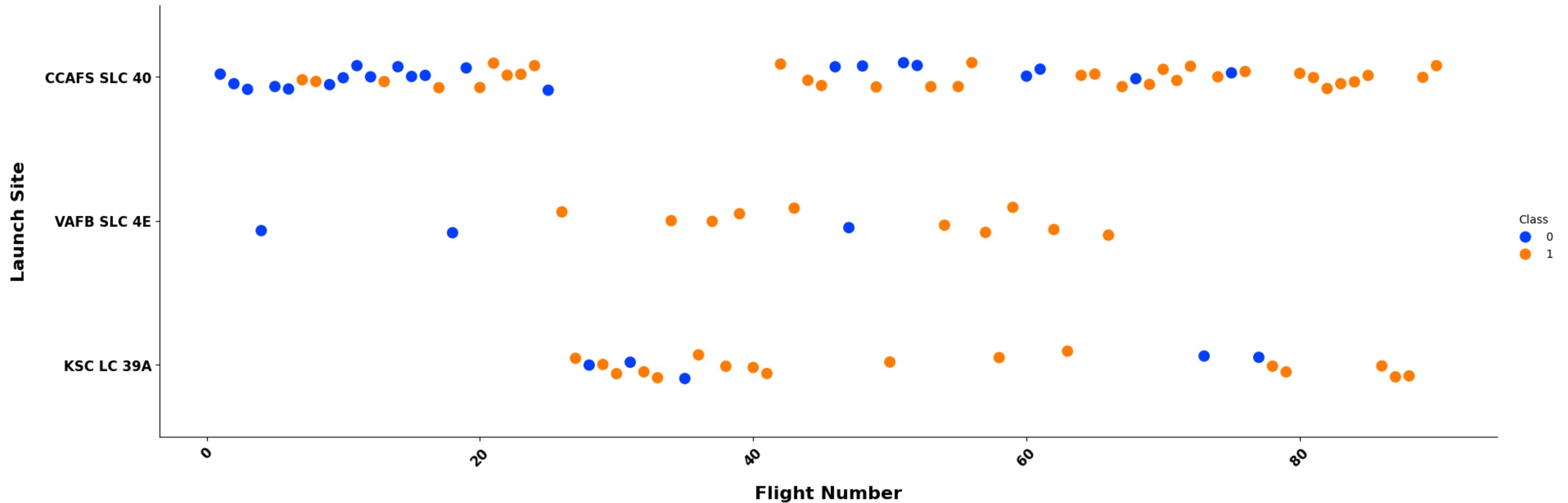
- **VISUALISATION**
- **SQL**



VISUALIZATION

FLIGHT NUMBER VS LAUNCH SITE

Flight Numbers by Launch Site and Class

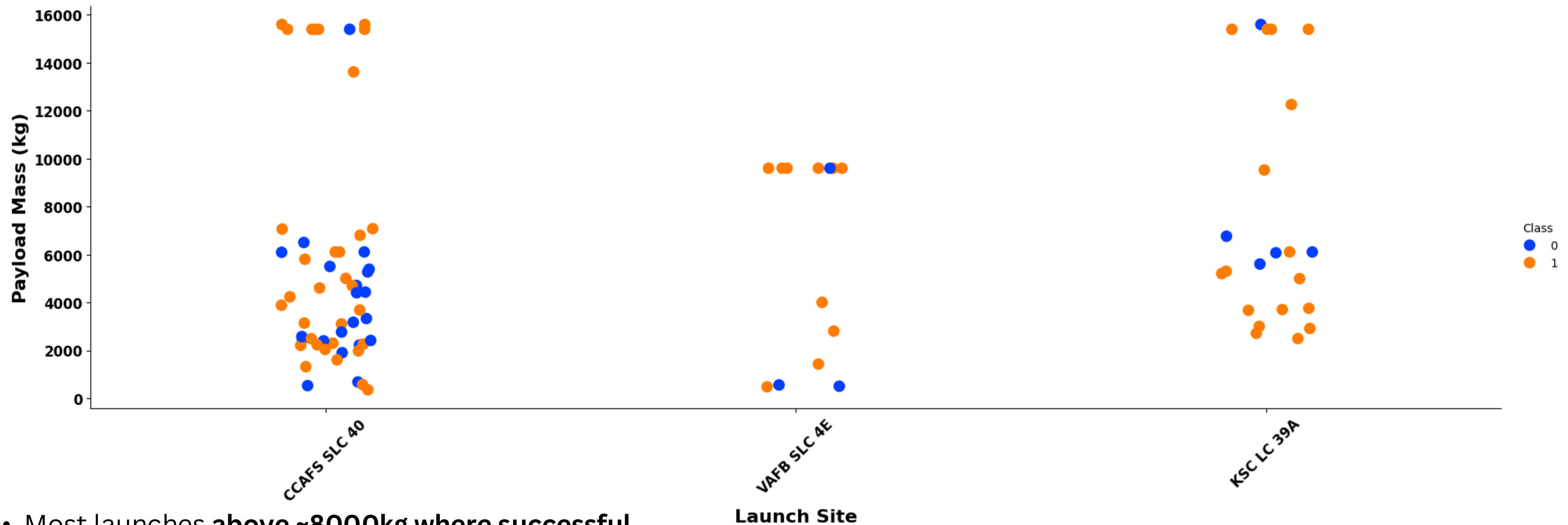


- Blue indicates failed launches and orange indicates successful launches.
- **CCAFS SLC 40** contains **highest volume of launches** although **not the most successful** launches compared to volume.
- **KSC LC 39A** is by far the **most successful** where only 5/22 launches being unsuccessful.
- Visualization suggests an **increase in success rate as number of flights increases**.

VISUALIZATION

PAYLOAD VS LAUNCH SITE

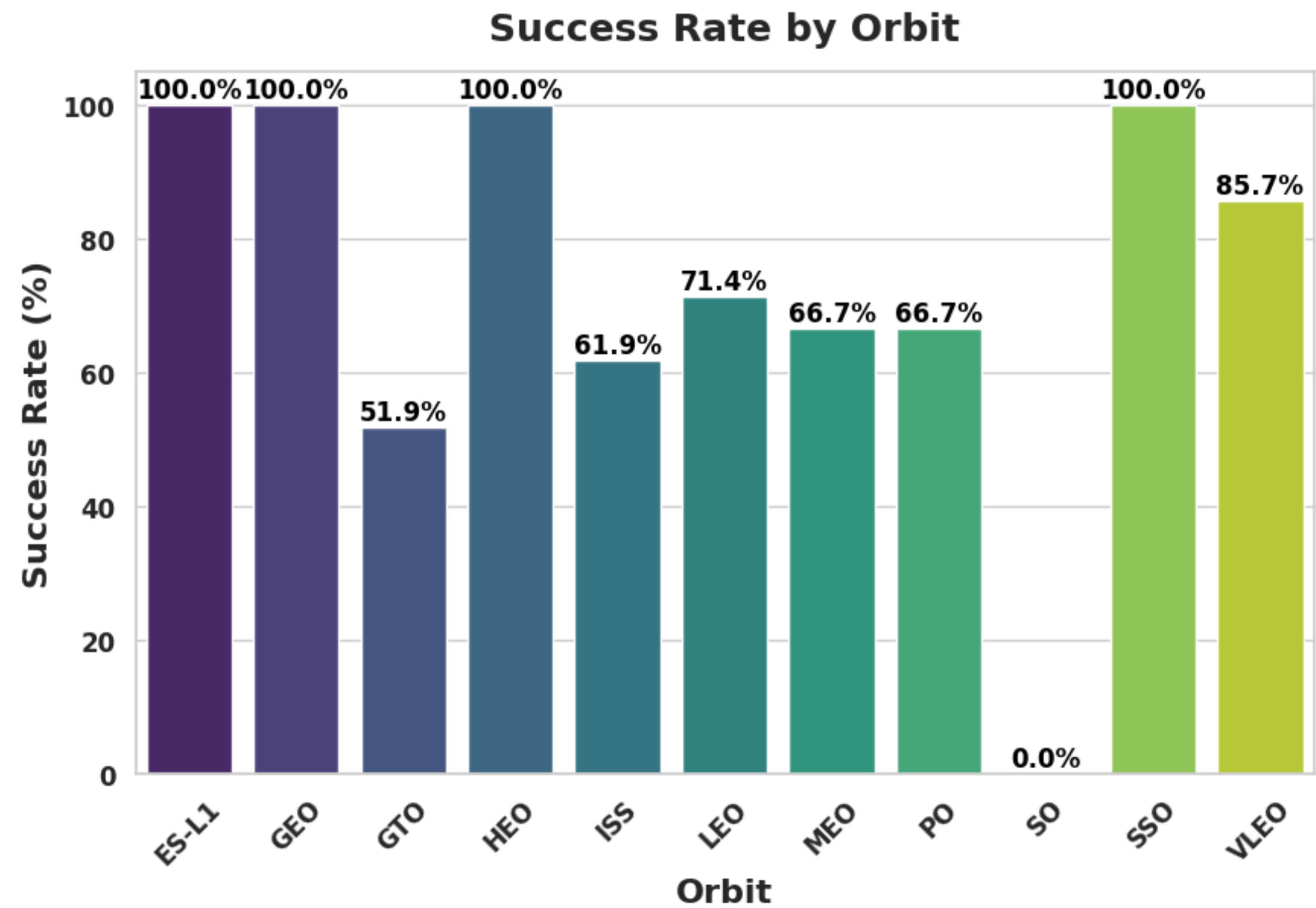
Payload Mass by Launch Site and Class



- Most launches **above ~8000kg** where **successful**
- **KSC LC 39A** has a **perfect success rate** for launches **under ~5000kg**. Between **~5500kg - ~8000kg** most of the launches are **unsuccessful**.
- **VAFB SKC 4E** did not launch anything **above 10 000kg** seeming that they found **success at 10 000kg** with only one failed launch.
- Thus, the **higher the payload mass, the higher the success rate**, excluding KSC LC 39A from this statement.

VISUALIZATION

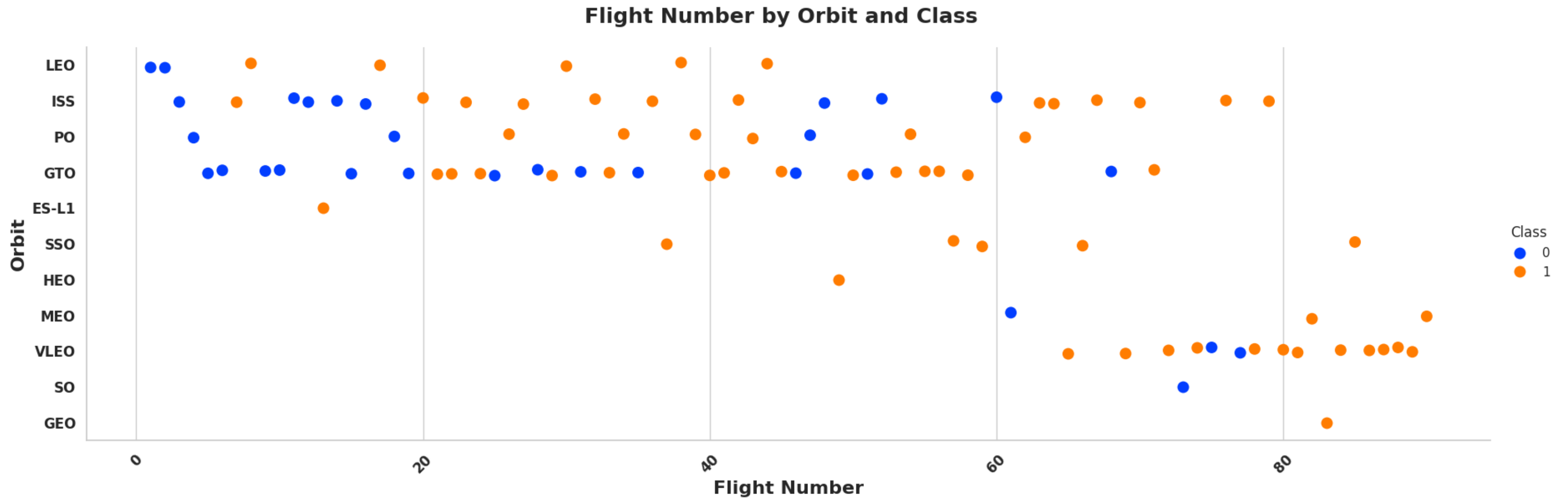
SUCCESS RATE VS ORBIT TYPE



- ES-L1, GEO, HEO, SSO 100% success rate.
- GTO, ISS, LEO, MEO, PO 50-80% success rate.
- SO 0% success rate

VISUALIZATION

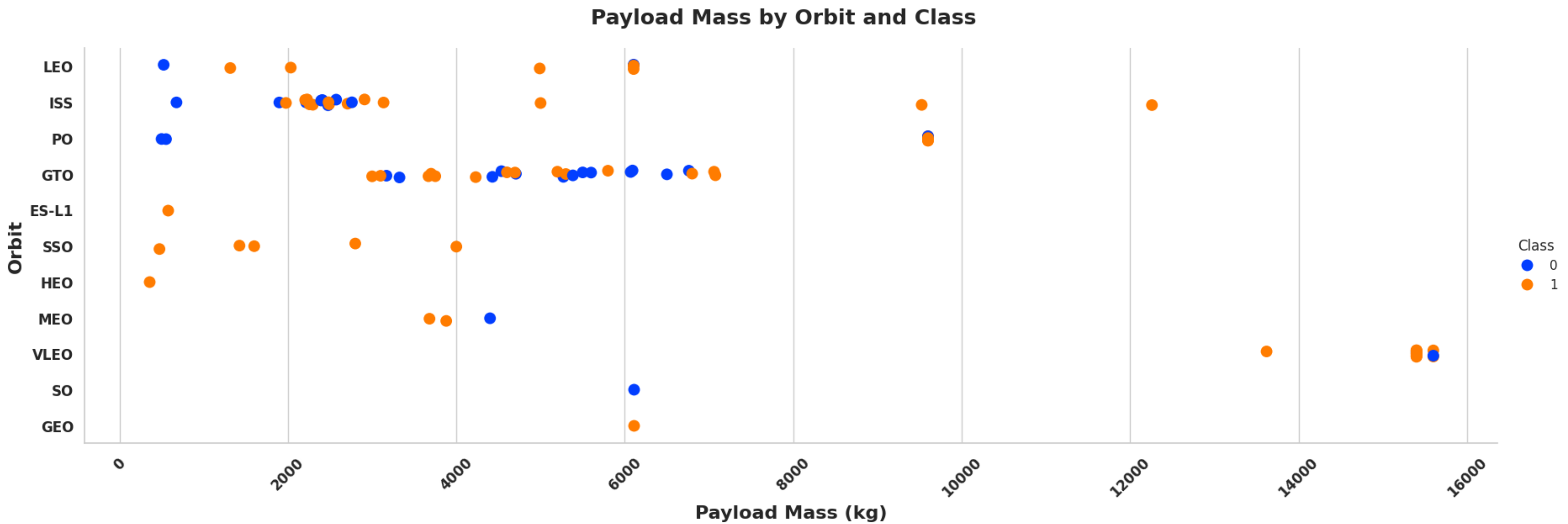
FLIGHT NUMBER VS ORBIT TYPE



- SpaceX started launches in **LEO orbits where moderate success** was found.
- Most **recent launches** are mostly **VLEO** orbit thus **SpaceX performs better in lower orbits**.
- It seems that **LEOs success related is to flight number** but GTO orbit breaks this relationship.
- There was **major success** between flight **20 and ~45** in orbits **LEO, ISS, PO**.

VISUALIZATION

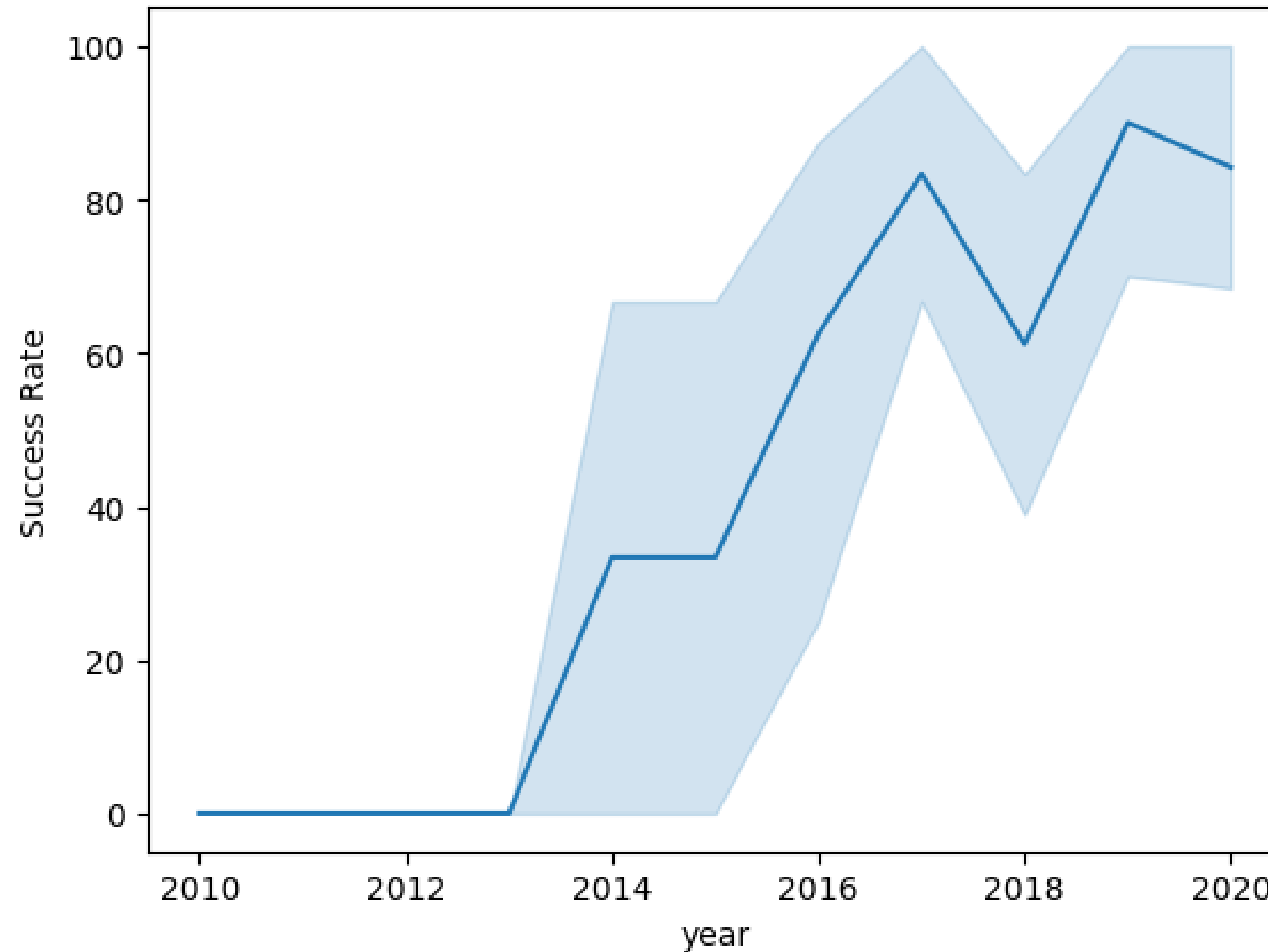
PAYLOAD MASS VS ORBIT TYPE



- LEO best performed with payload mass above ~1500kg.
- Heavy payloads are better with LEO, ISS and PO orbits.
- The GTO orbit has mixed success with heavier payloads thus a correlation cannot be made between payload mass and orbit type.

VISUALIZATION

LAUNCH SUCCESS YEARLY TREND



- Success rate **improved** from 2013-2017 and 2018-2019.
- Success rate **decreased** from 2017-2018 and 2019-2020.
- Overall the success rate improved from 2013.

SQL

ALL LAUNCH SITE NAMES

```
%sql select DISTINCT Launch_Site from SPACEXTBL
```

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

- Using DISTINCT in the query allows to show unique values in the Launch_Site column from the database.
- Research about CCAF LC-40 and CCAFS SLC-40 concluded that they are the same launch site in Florida operated by different hosts, first the United states air force and then SpaceX.
https://en.wikipedia.org/wiki/Cape_Canaveral_Space_Launch_Complex_40



SQL

LAUNCH SITE NAMES BEGIN WITH ‘CCA’

```
%sql SELECT * from SPACEXTBL where Launch_Site like 'CCA%' limit 5
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- First 5 entries in database with launch site name beginning with CCA. % - include anything after CCA.

SQL

TOTAL PAYLOAD MASS FROM NASA

```
%sql SELECT sum(PAYLOAD_MASS_KG_) FROM SPACEXTBL where customer LIKE 'NASA (CRS)'
```

sum(PAYLOAD_MASS_KG_)

45596

- Total payload mass where NASA was a customer.



SQL

AVERAGE PAYLOAD MASS BY F9 V1.1

```
%sql SELECT avg(PAYLOAD_MASS_KG_) FROM SPACEXTBL where Booster_Version LIKE 'F9 v1.1%'
```

avg(PAYLOAD_MASS_KG_)

2534.6666666666665

- Average payload mass of launches which used booster version F9 v1.1
- **Average payload mass is very low.**



SQL

FIRST SUCCESSFUL GROUND PAD LANDING DATE

```
%sql SELECT min(Date) FROM SPACEXTBL WHERE landing_Outcome like 'Success (ground pad)'
```

min(Date)

2015-12-22

- First successful ground pad landing date.
- **Successful landings** generally appear starting from **2014** as also indicated by the **launch success vs year line graph**.



SQL

SUCCESSFUL DRONE SHIP LANDING WITH PAYLOAD BETWEEN 4000 AND 6000

```
%sql SELECT Booster_Version FROM SPACEXTBL WHERE (Mission_Outcome like 'Success') AND (PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000) AND (Landing_Outcome like 'Success (drone ship)')
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- Four booster versions that had successful drone ship landings and a payload mass between 4000 and 6000.



TOTAL NUMBER OF EACH MISSION OUTCOME (SUCCESS & FAILURE)

```
%sql SELECT Mission_Outcome, count(*) as Count FROM SPACEXTBL GROUP by Mission_Outcome ORDER BY Mission_Outcome
```

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- 99 Success
- 1 Failure
- 1 Success (payload status unclear)



SQL

BOOSTERS THAT CARRIED MAXIMUM PAYLOAD

```
maxm = %sql SELECT max(PAYLOAD_MASS__KG_) FROM SPACEXTBL
maxv = maxm[0][0]
%sql SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_=(select max(PAYLOAD_MASS__KG_) from SPACEXTBL)
```

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

- Booster versions that carried the **highest payload mass of 15600kg**.
- All the boosters are of the **F9 B% B10xx.x** variety.
- This indicates that **payload mass correlates with the booster version** that is used.



SQL

2015 FAILED DRONE SHIP LANDING RECORDS

```
%%sql
SELECT
CASE
WHEN substr(Date, 6, 2) = '01' THEN 'January'
WHEN substr(Date, 6, 2) = '02' THEN 'February'
WHEN substr(Date, 6, 2) = '03' THEN 'March'
WHEN substr(Date, 6, 2) = '04' THEN 'April'
WHEN substr(Date, 6, 2) = '05' THEN 'May'
WHEN substr(Date, 6, 2) = '06' THEN 'June'
WHEN substr(Date, 6, 2) = '07' THEN 'July'
WHEN substr(Date, 6, 2) = '08' THEN 'August'
WHEN substr(Date, 6, 2) = '09' THEN 'September'
WHEN substr(Date, 6, 2) = '10' THEN 'October'
WHEN substr(Date, 6, 2) = '11' THEN 'November'
WHEN substr(Date, 6, 2) = '12' THEN 'December'
END AS Month,
landing_outcome,
booster_version,
launch_site
FROM SPACEXTABLE
WHERE
landing_outcome = 'Failure (drone ship)' AND
substr(Date, 1, 4) = '2015';
```

Month	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- 2 occurrences where stage 1 of the launch failed.
- Both are CCAFS LC-40 which is the 2nd most successful launch site.
- They used the same booster version but earlier models, this is likely why they failed.



SQL

RANK LANDING OUTCOMES BETWEEN 2010-06-04 AND 2017-03-20

```
%%sql
SELECT
    landing_outcome,
    COUNT(*) AS outcome_count
FROM SPACEXTABLE
WHERE
    Date >= '2010-06-04' AND
    Date <= '2017-03-20'
GROUP BY
    landing_outcome
ORDER BY
    outcome_count DESC;
```

Landing_Outcome	outcome_count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- Landing outcomes between above dates.
- **Success type landings**
 - drone ship
 - ground pad
 - controlled (ocean) -can be seen as fail
- **Failed landings/launch attempts**
 - no attempt
 - drone ship
 - uncontrolled (ocean)
 - failure (parachute)
 - Precluded (drone ship)
- **There were 8 successful landings during the time period.**

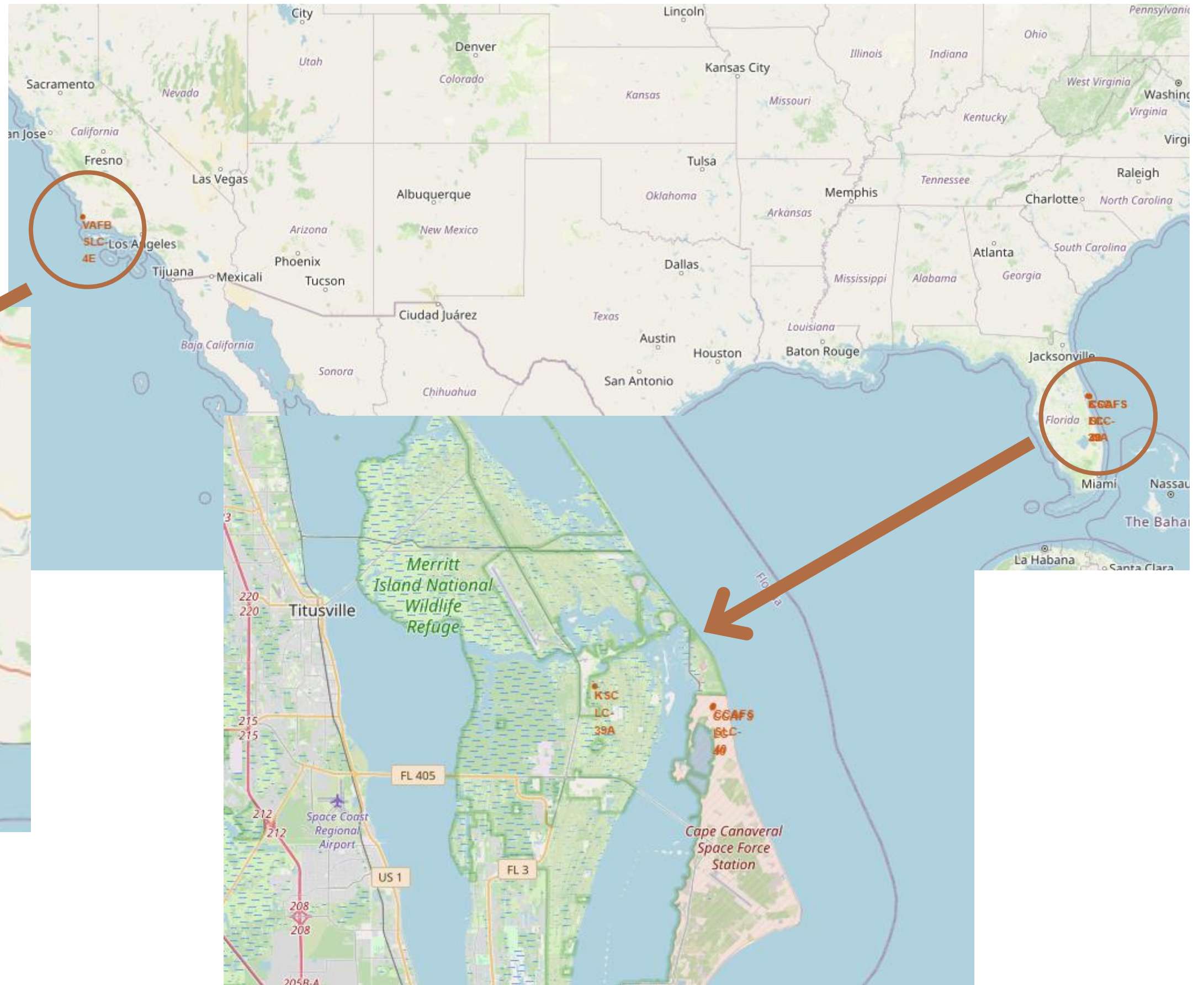


LAUNCH SITES PROXIMITIES ANALYSIS



LAUNCH SITE LOCATIONS

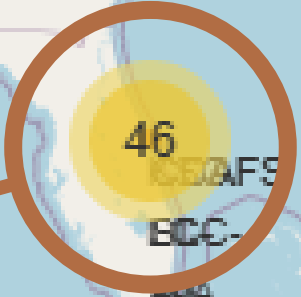
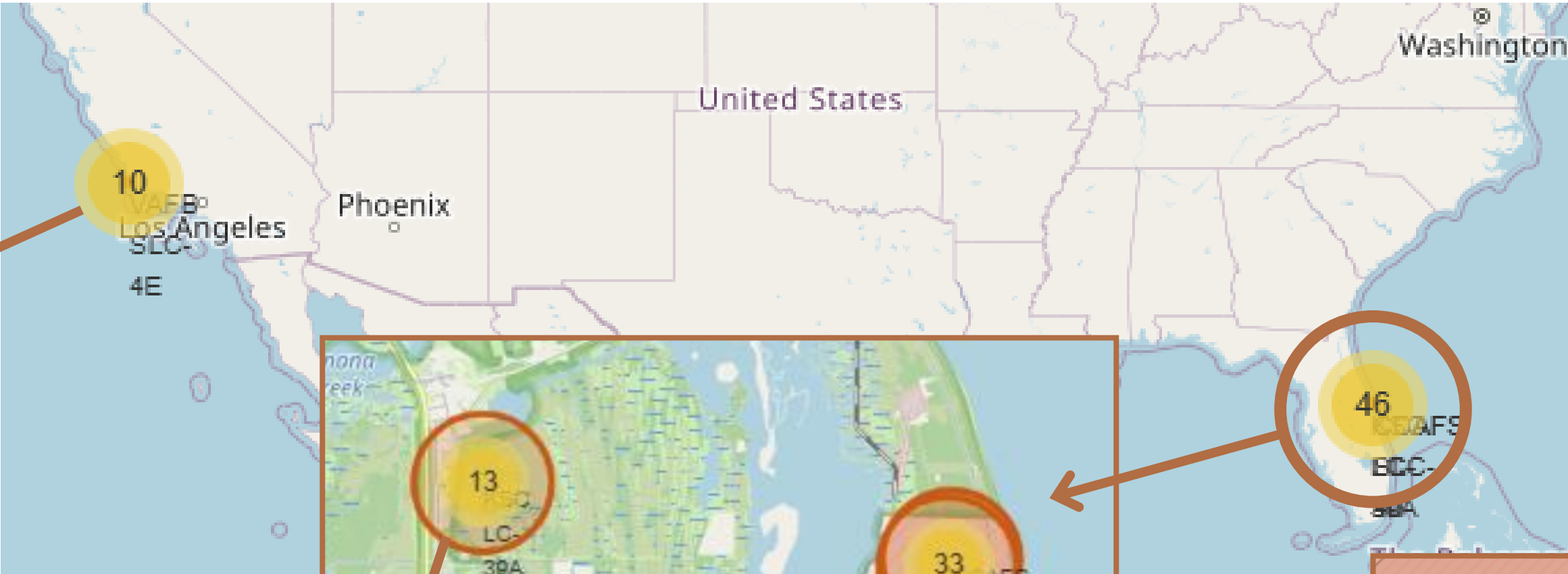
- SpaceX launch sites:
 - Florida and California



LAUNCH MARKERS



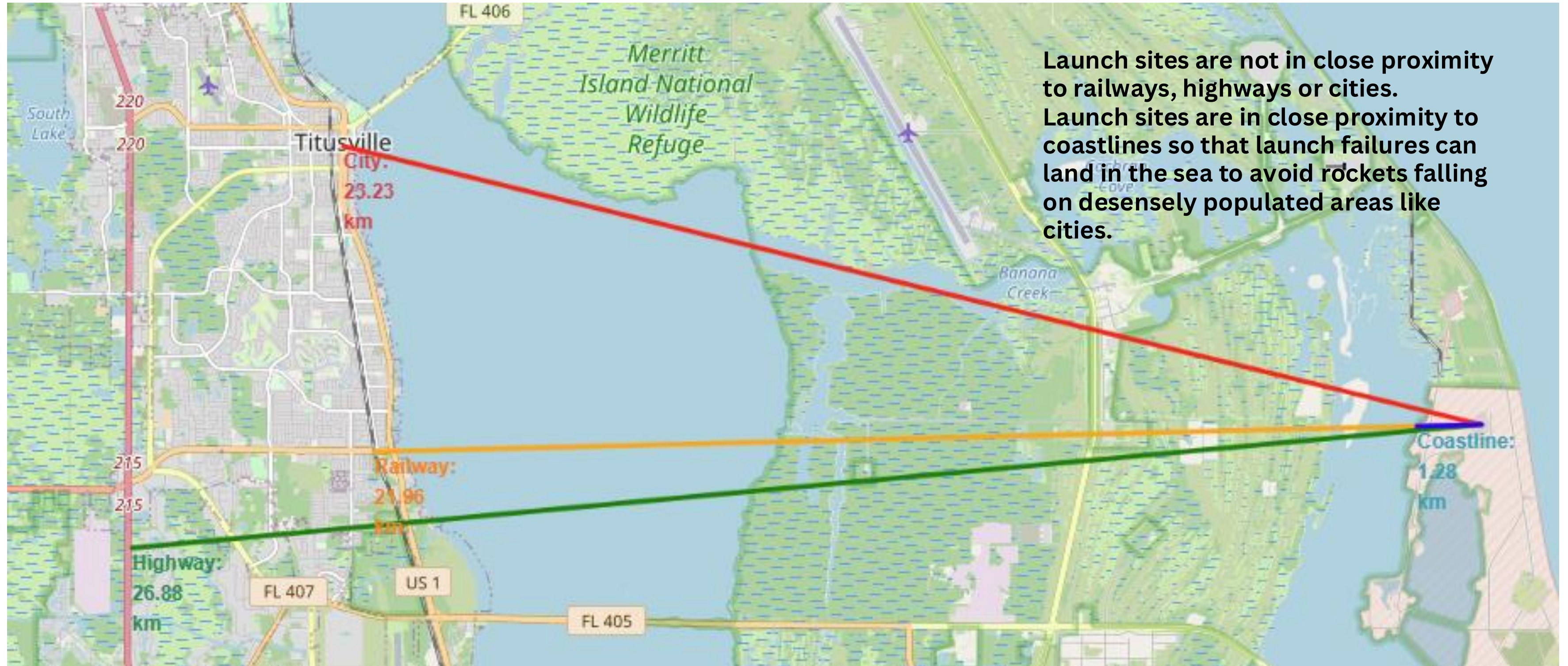
California Lunch Sites
Green - successful
Red - failures
Clusters can be clicked on to display each landing



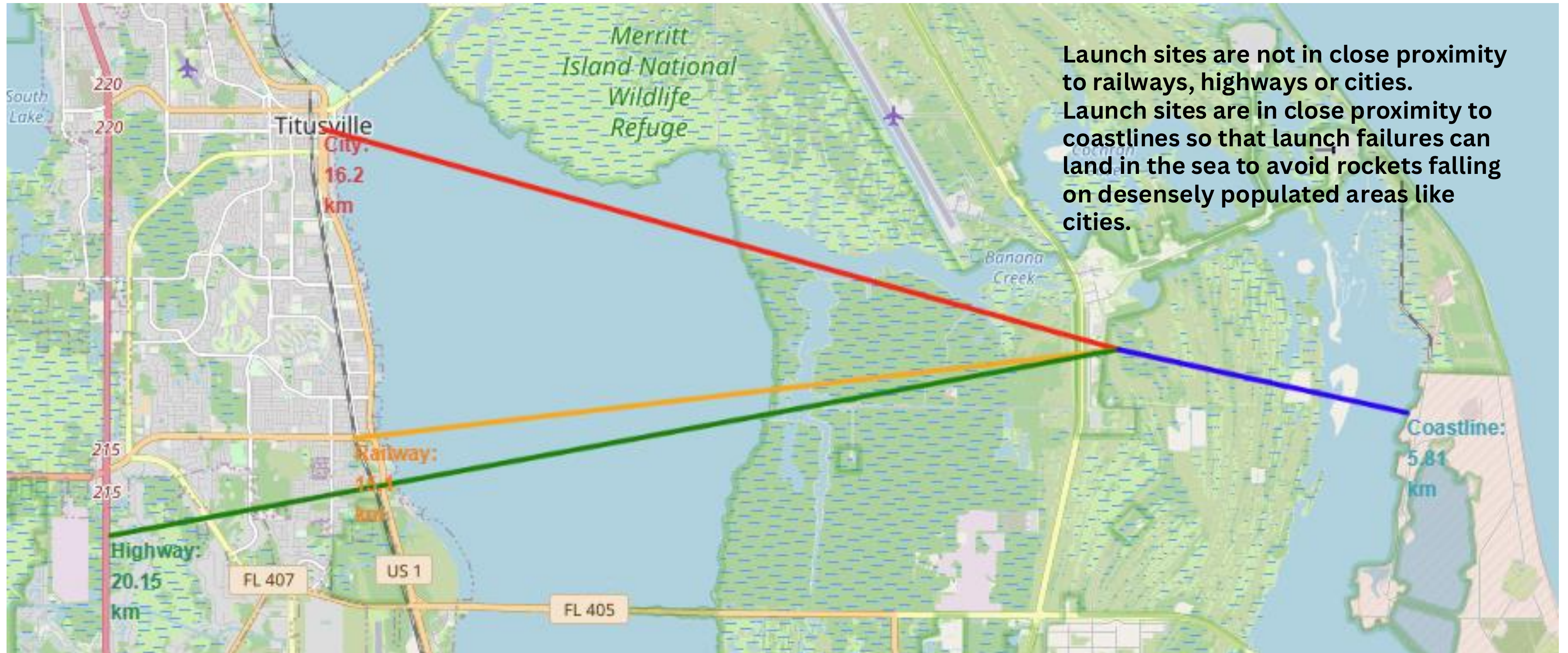
Florida Launch Sites
Green - successful
Red - failures



CCAFS SLC-40 & CCAFS LC-40 PROXIMITIES



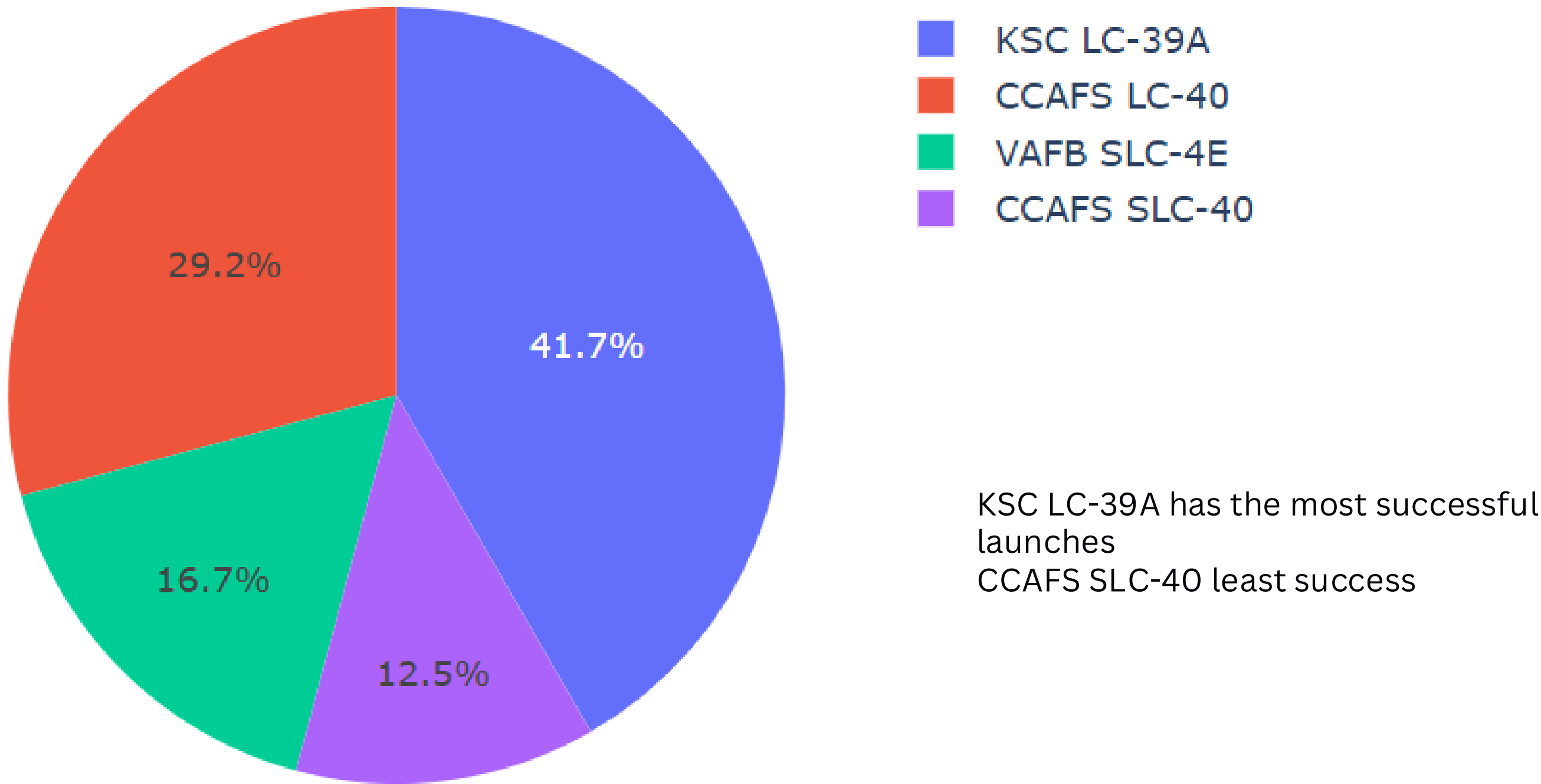
KSC LC-39A PROXIMITIES



BUILD A DASHBOARD WITH PLOTLY DASH

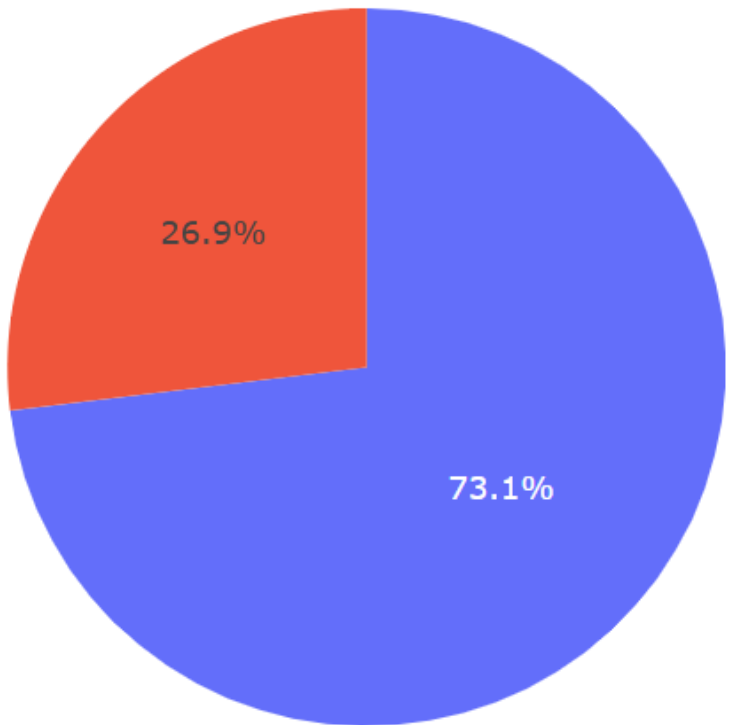


TOTAL SUCCESS LAUNCHES BY SITE

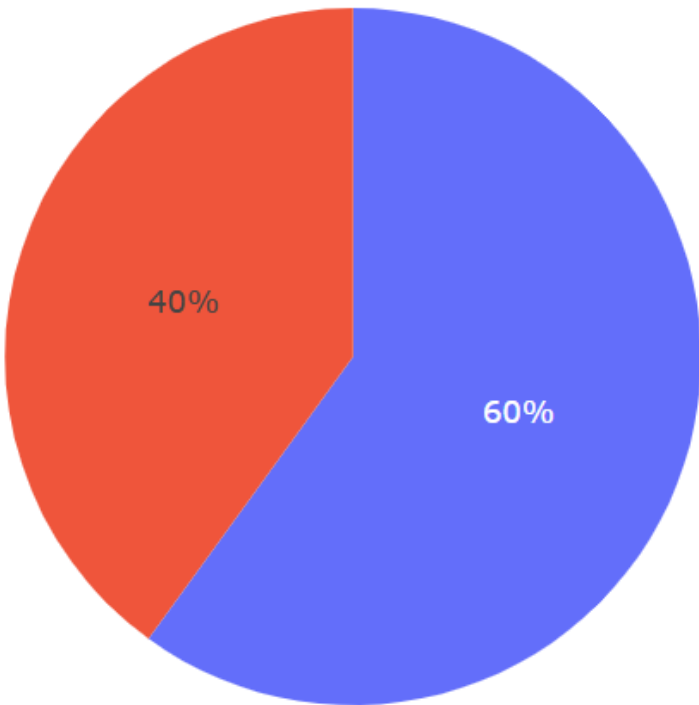


TOTAL SUCCESS LAUNCHES

CCAFS LC-40

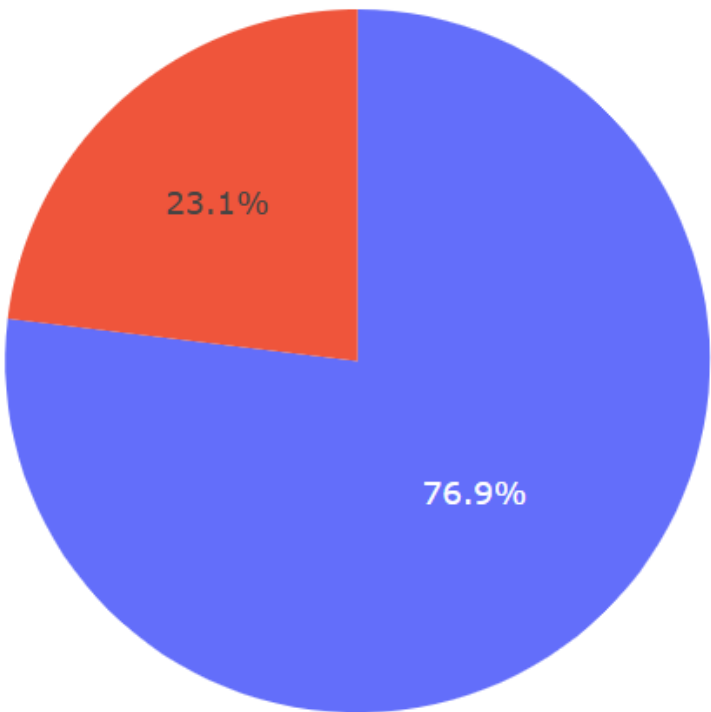


VAFB SLC-4E

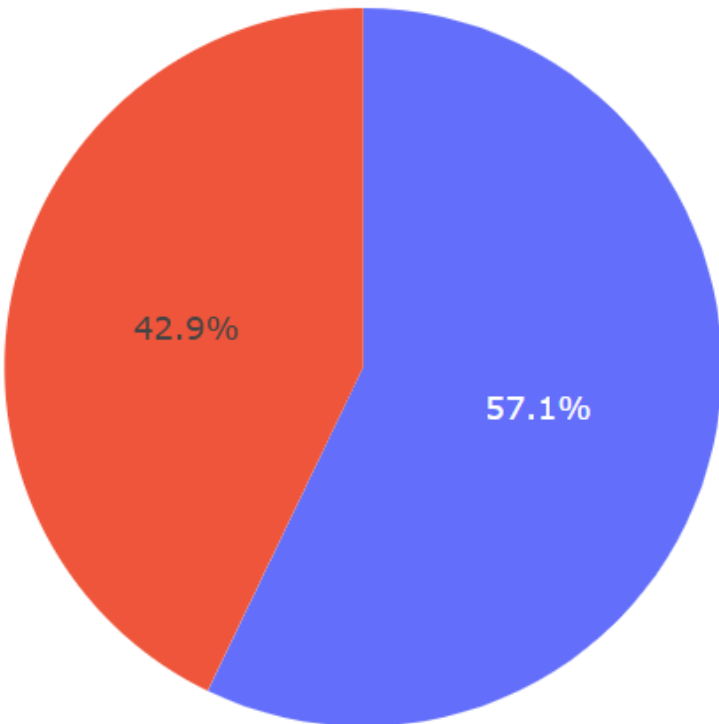


Success
Failure

KSC LC-39A



VAFB SLC-4E



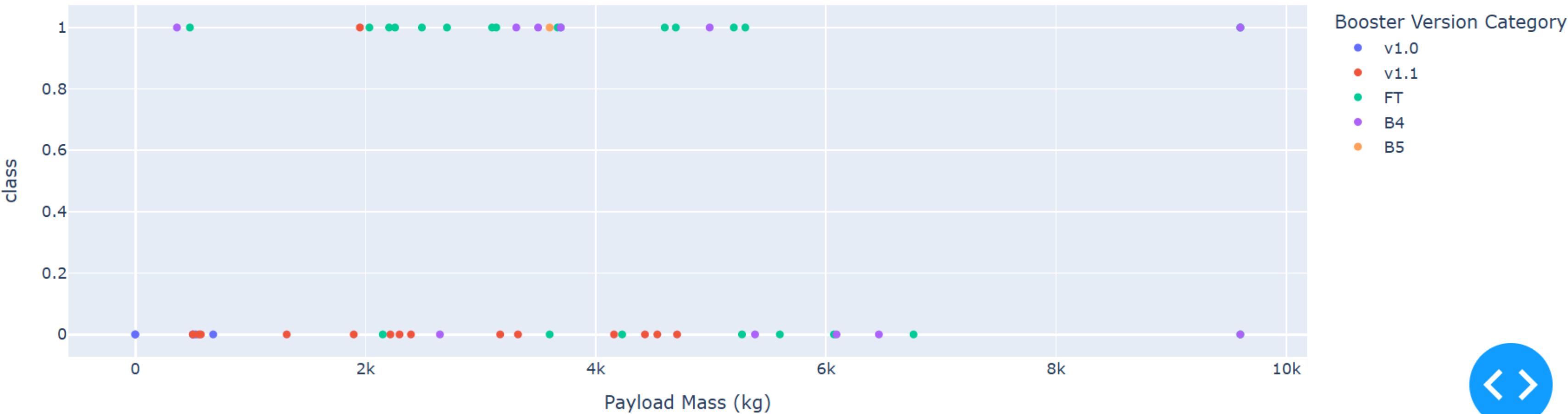
KSC LC-39A has
the most
successful
launches

PAYLOAD VS OUTCOME FOR ALL SITES

Payload range (Kg):



Payload vs. Outcome for All Sites

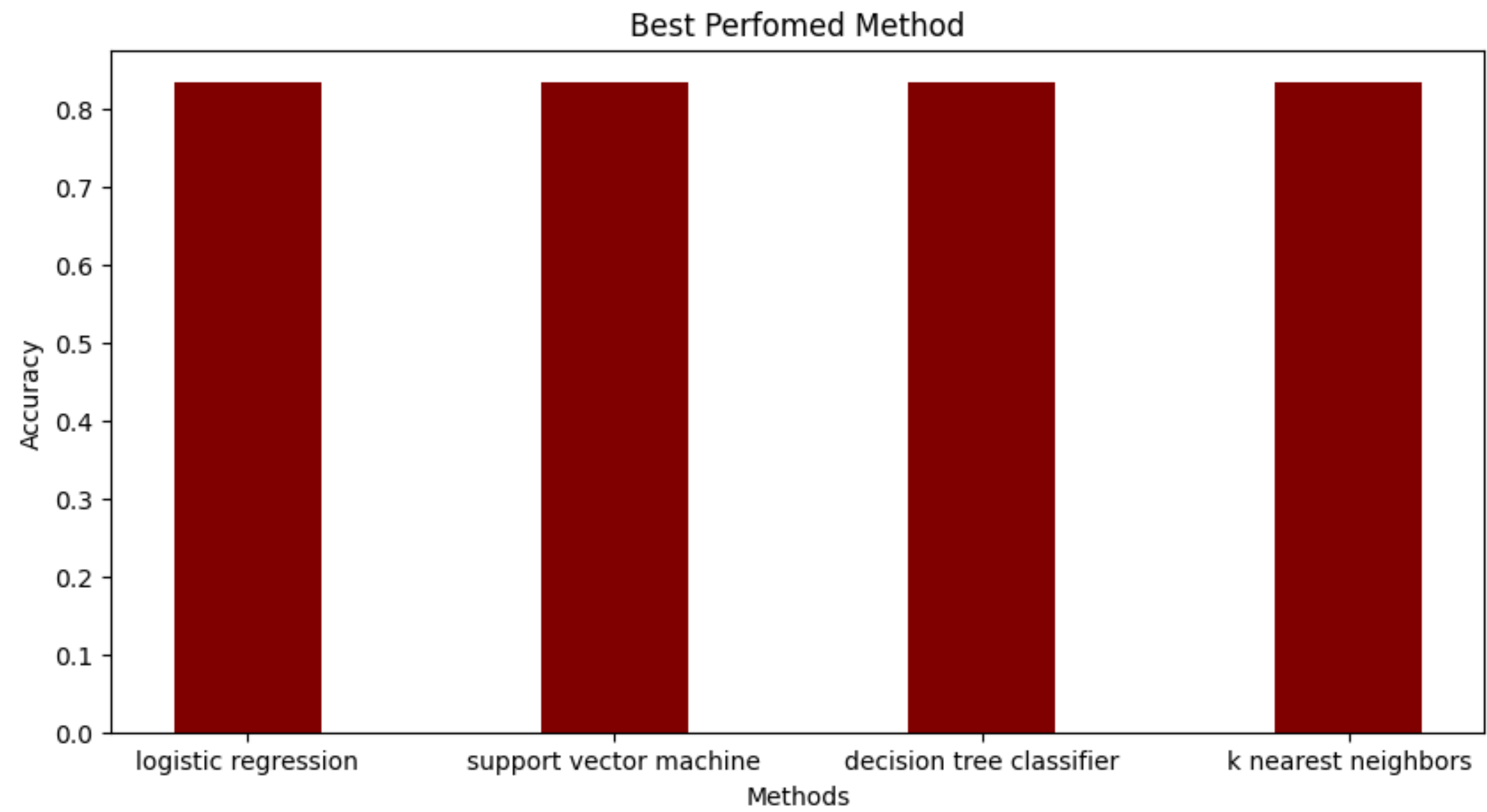


- More successful landings with payload mass of less than 5500kg.
- Most successful boosters include FT and B4.
- Boosters v1.0 and v1.1 doesn't show positive results.
- Class 1 - successful landing
- Class 0 - failure
- Booster versions shown in different colours

PREDICTIVE ANALYSIS (CLASSIFICATION)

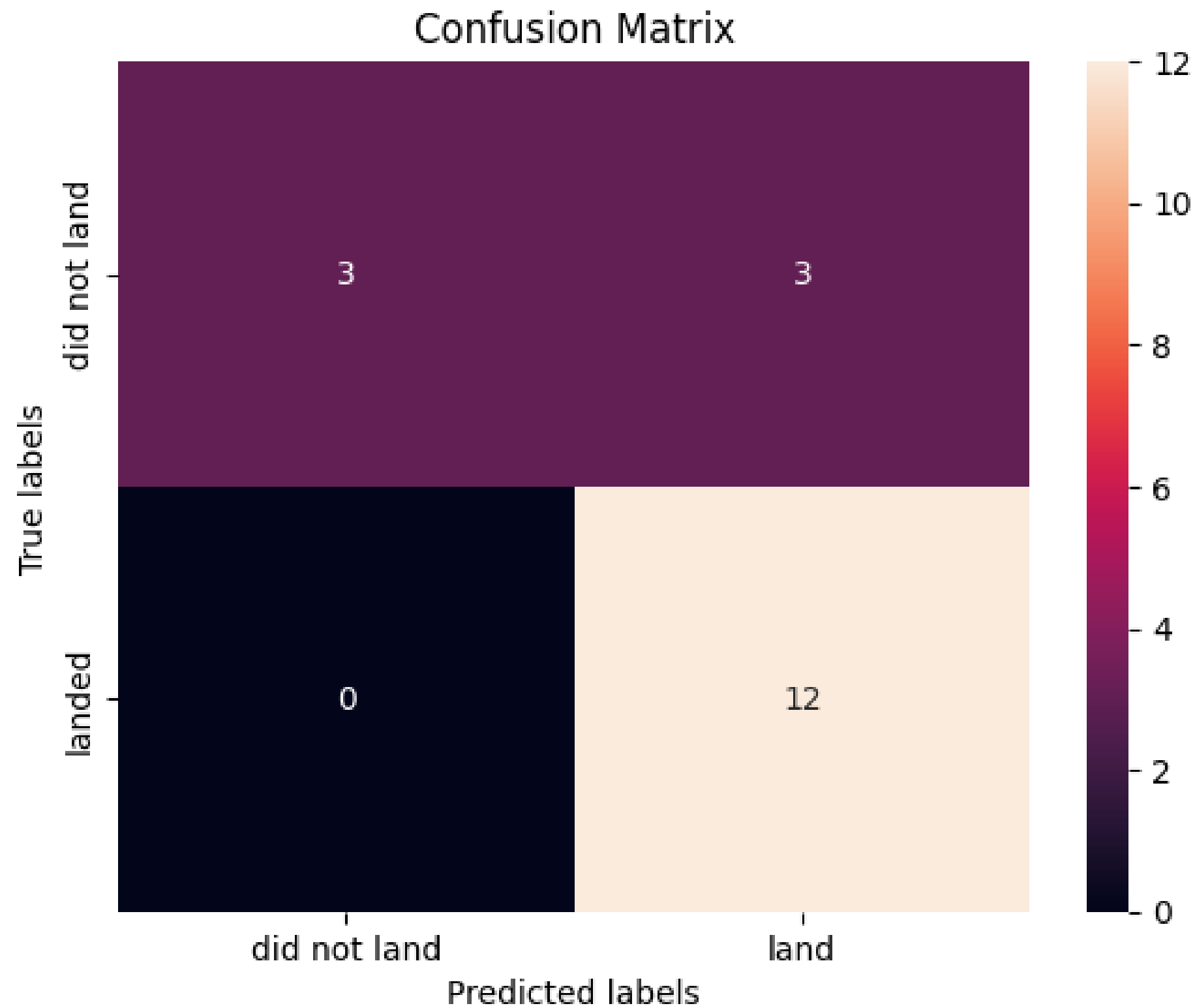


CLASSIFICATION ACCURACY



- All models had the same accuracy if 83.33%
- We likely need more data to determine the best model.
- Looking at best performed method, the decision tree classifier is the best if we look down to the decimal.

CONFUSION MATRIX



- The model predicts 12 successful landings
- Predicts 3 unsuccessful landings.
- Predicts 3 successful landing (false positive)
- Thus our model over predicts successful landings and our major problem is false positives.

CONCLUSION

- We developed a machine learning model for Space Y to predict whether the Falcon 9 first stage will land successfully based on historical data and launch parameters.
- Used SpaceX API and Web Scraping techniques to retrieve data.
- Wrangled data to get needed sources and used data visualization and SQL to retrieve needed information and create relevant graphs to conclude on relationships between parameters.
- Machine learning model created with 83.33% accuracy.
- The decision tree classifier is the best for machine learning for this dataset.
- KSC LC -39A had the most successful launches out of all the sites.
- Orbit ES-L1, GEO, HEO, SSO has the best success rate.\
- Space Y can predict with a relevantly high accuracy whether a launch will have a successful stage 1 landing, to determine whether the launch should be made or not.
- More data should be collected to better determine the best model for machine learning model to improve accuracy

APPENDIX

Github repository url:

<https://github.com/Ankedefrey/Data-Science-Capstone-Project>